

Artwork: Test Image1019

Scale-Invariant Coordinate System

$$\tilde{X} = 2 \left(\frac{x}{W} \right) - 1, \quad \tilde{Y} = 1 - 2 \left(\frac{y}{H} \right) X = \frac{\left(\tilde{X} + \frac{1}{8} \right)}{\frac{37}{45}}, \quad Y = \frac{\left(\tilde{Y} + \frac{6}{85} \right)}{\frac{151}{170}}, \quad \theta = \arctan \left(\frac{Y}{X} \right)$$

Layer Definitions Boundary Manifolds

$$C_0(X, Y, v) = \frac{23 - v}{40} \cdot \left(1 - \frac{52}{159} X - \frac{2}{85} Y \right)$$

$$X_1 = \left(X + \frac{4}{61} \right), \quad Y_1 = \left(Y + \frac{64}{77} \right)$$

$$\tau_1(X, Y) = -\frac{4}{123} \cos^4 \left(\frac{531}{95} X_1 + \frac{2698}{139} Y_1 - \frac{53}{84} \cos \left(\frac{643}{195} X_1 + \frac{747}{197} Y_1 \right) \right) - \frac{8}{155} \cos^6 \left(\frac{2089}{76} X_1 + \frac{2229}{199} Y_1 - \frac{18}{13} \cos \left(\frac{736}{101} X_1 + \frac{632}{193} Y_1 \right) \right) - \frac{1}{57} \cos^8 \left(\frac{2271}{190} X_1 + \frac{6459}{182} Y_1 - \frac{5}{1} \cos \left(\frac{2271}{190} X_1 + \frac{6459}{182} Y_1 \right) \right)$$

$$X_2 = \left(X - \frac{5}{38} \right), \quad Y_2 = \left(Y + \frac{5}{107} \right)$$

$$\tau_2(X, Y) = -\frac{38}{193} \cos^2 (16X_2 + 6Y_2) - \frac{22}{111} \cos^4 (6X_2 + 20Y_2) - \frac{37}{189} \cos^6 (28X_2 + 12Y_2) - \frac{37}{188} \cos^8 (12X_2 + 36Y_2) - \frac{32}{165} \cos^{12} (48X_2 + 20Y_2) - \frac{21}{107} \cos^{16} (28X_2 + 12Y_2)$$

$$X_3 = \left(X + \frac{34}{161} \right), \quad Y_3 = \left(Y - \frac{1}{91} \right)$$

$$\tau_3(X, Y) = -\frac{1}{156} \cos^2 \left(\frac{1780}{113} X_3 + \frac{553}{97} Y_3 + \frac{9}{58} \cos \left(\frac{199}{51} X_3 + \frac{347}{186} Y_3 \right) \right) - \frac{20}{163} \cos^4 \left(\frac{193}{31} X_3 + \frac{2870}{143} Y_3 \right) - \frac{48}{179} \cos^6 \left(\frac{4235}{151} X_3 + \frac{1559}{131} Y_3 - \frac{55}{161} \cos \left(\frac{989}{171} X_3 + \frac{474}{163} Y_3 \right) \right)$$

$$X_4 = \left(X - \frac{17}{174} \right), \quad Y_4 = \left(Y - \frac{2}{191} \right)$$

$$\tau_4(X, Y) = -\frac{76}{161} \cos^2 \left(\frac{2082}{133} X_4 + \frac{1121}{192} Y_4 + \frac{1}{2} \cos \left(\frac{596}{179} X_4 + \frac{219}{196} Y_4 \right) \right) - \frac{19}{49} \cos^4 \left(\frac{592}{95} X_4 + \frac{3577}{174} Y_4 - \frac{4}{31} \cos \left(\frac{264}{97} X_4 + \frac{988}{179} Y_4 \right) \right) - \frac{62}{181} \cos^6 \left(28X_4 + \frac{2035}{179} Y_4 - \frac{101}{100} \cos \left(\frac{264}{97} X_4 + \frac{988}{179} Y_4 \right) \right)$$

$$X_5 = \left(X - \frac{8}{161} \right), \quad Y_5 = \left(Y + \frac{2}{37} \right)$$

$$\tau_5(X, Y) = -\frac{32}{187} \cos^2 \left(\frac{684}{43} X_5 + \frac{910}{157} Y_5 + \frac{2}{27} \cos \left(\frac{223}{56} X_5 + 2Y_5 \right) \right) + \frac{9}{58} \cos^4 \left(\frac{1111}{189} X_5 + \frac{3694}{185} Y_5 + \frac{34}{143} \cos \left(\frac{401}{191} X_5 + \frac{947}{188} Y_5 \right) \right) - \frac{49}{183} \cos^6 \left(\frac{5041}{180} X_5 + \frac{2197}{181} Y_5 + \frac{21}{157} \cos \left(\frac{401}{191} X_5 + \frac{947}{188} Y_5 \right) \right)$$

$$X_6 = \left(X - \frac{23}{116}\right), \quad Y_6 = \left(Y + \frac{25}{169}\right)$$

$$\tau_6(X, Y) = -\frac{53}{150} \cos^2 \left(\frac{2705}{166} X_6 + \frac{874}{143} Y_6 - \frac{185}{197} \cos \left(\frac{850}{191} X_6 + \frac{142}{115} Y_6 \right) \right) - \frac{27}{154} \cos^4 \left(\frac{569}{91} X_6 + \frac{2857}{141} Y_6 - \frac{1}{2} \cos \left(\frac{327}{110} X_6 + \frac{59}{13} Y_6 \right) \right) - \frac{43}{195} \cos^6 \left(\frac{4174}{149} X_6 + \frac{1453}{126} Y_6 - \frac{6}{25} \cos \left(\frac{109}{110} X_6 + \frac{109}{110} Y_6 \right) \right)$$

$$X_7 = \left(X - \frac{40}{199}\right), \quad Y_7 = \left(Y + \frac{7}{190}\right)$$

$$\tau_7(X, Y) = \frac{3}{125} \cos^2 \left(\frac{2558}{163} X_7 + \frac{1256}{197} Y_7 - \frac{119}{104} \cos \left(\frac{498}{179} X_7 + \frac{247}{125} Y_7 \right) \right) + \frac{1}{86} \cos^6 \left(\frac{4181}{150} X_7 + \frac{1791}{163} Y_7 + \frac{127}{183} \cos \left(\frac{125}{22} X_7 + \frac{236}{101} Y_7 \right) \right) - \frac{1}{131} \cos^{12} \left(\frac{7673}{163} X_7 + \frac{3719}{185} Y_7 + \frac{1}{2} \cos \left(\frac{109}{110} X_7 + \frac{109}{110} Y_7 \right) \right)$$

$$X_8 = \left(X - \frac{27}{185}\right), \quad Y_8 = \left(Y - \frac{11}{49}\right)$$

$$\tau_8(X, Y) = -\frac{40}{133} \cos^2 \left(\frac{2778}{181} X_8 + \frac{981}{191} Y_8 + \frac{279}{175} \cos \left(\frac{627}{155} X_8 + \frac{239}{189} Y_8 \right) \right) - \frac{13}{198} \cos^4 \left(\frac{902}{195} X_8 + \frac{2998}{151} Y_8 + \frac{190}{173} \cos \left(\frac{14}{5} X_8 + \frac{442}{79} Y_8 \right) \right) - \frac{3}{154} \cos^6 \left(\frac{3828}{139} X_8 + \frac{1944}{167} Y_8 + \frac{1}{2} \cos \left(\frac{109}{110} X_8 + \frac{109}{110} Y_8 \right) \right)$$

$$X_9 = \left(X + \frac{9}{56}\right), \quad Y_9 = \left(Y - \frac{123}{149}\right)$$

$$\tau_9(X, Y) = \frac{16}{199} \cos^4 \left(\frac{1051}{155} X_9 + \frac{3581}{175} Y_9 - \frac{8}{61} \cos \left(\frac{99}{70} X_9 + \frac{697}{156} Y_9 \right) \right) - \frac{2}{131} \cos^8 \left(\frac{1993}{162} X_9 + \frac{5121}{142} Y_9 \right) - \frac{1}{128} \cos^{12} \left(\frac{9246}{191} X_9 + \frac{3881}{197} Y_9 - \frac{8}{23} \cos \left(\frac{1150}{159} X_9 + \frac{53}{14} Y_9 \right) \right)$$

$$X_{10} = \left(X + \frac{28}{99}\right), \quad Y_{10} = \left(Y - \frac{15}{77}\right)$$

$$\tau_{10}(X, Y) = -\frac{107}{147} \cos^2 \left(\frac{2755}{163} X_{10} + \frac{710}{141} Y_{10} + \frac{207}{191} \cos \left(\frac{887}{187} X_{10} + \frac{144}{149} Y_{10} \right) \right) - \frac{90}{137} \cos^4 \left(\frac{981}{193} X_{10} + \frac{3406}{163} Y_{10} + \frac{163}{196} \cos \left(\frac{143}{127} X_{10} + \frac{949}{158} Y_{10} \right) \right) - \frac{19}{135} \cos^6 \left(\frac{4825}{171} X_{10} + \frac{1}{2} \cos \left(\frac{109}{110} X_{10} + \frac{109}{110} Y_{10} \right) \right)$$

$$X_{11} = \left(X + \frac{28}{87}\right), \quad Y_{11} = \left(Y + \frac{13}{192}\right)$$

$$\tau_{11}(X, Y) = -\frac{10}{83} \cos^2 \left(\frac{2818}{179} X_{11} + \frac{82}{13} Y_{11} - \frac{25}{87} \cos \left(\frac{191}{49} X_{11} + \frac{87}{40} Y_{11} \right) \right) - \frac{8}{89} \cos^4 \left(\frac{766}{123} X_{11} + \frac{816}{41} Y_{11} + \frac{41}{178} \cos \left(\frac{139}{79} X_{11} + \frac{1003}{195} Y_{11} \right) \right) - \frac{17}{141} \cos^6 \left(\frac{4111}{146} X_{11} + \frac{1980}{163} Y_{11} + \frac{1}{2} \cos \left(\frac{109}{110} X_{11} + \frac{109}{110} Y_{11} \right) \right)$$

$$X_{12} = \left(X + \frac{19}{70}\right), \quad Y_{12} = \left(Y - \frac{60}{191}\right)$$

$$\tau_{12}(X, Y) = -\frac{29}{194} \cos^2 \left(\frac{3028}{187} X_{12} + \frac{425}{66} Y_{12} + \frac{39}{34} \cos \left(\frac{383}{109} X_{12} + \frac{112}{95} Y_{12} \right) \right) - \frac{49}{171} \cos^4 \left(\frac{463}{94} X_{12} + \frac{1679}{86} Y_{12} + \frac{167}{165} \cos \left(\frac{303}{145} X_{12} + \frac{1033}{175} Y_{12} \right) \right) + \frac{9}{199} \cos^6 \left(\frac{5113}{186} X_{12} + \frac{1}{2} \cos \left(\frac{109}{110} X_{12} + \frac{109}{110} Y_{12} \right) \right)$$

$$X_{13} = \left(X + \frac{69}{154}\right), \quad Y_{13} = \left(Y - \frac{40}{199}\right)$$

$$\tau_{13}(X, Y) = \frac{35}{72} \cos^2 \left(\frac{2281}{141} X_{13} + \frac{967}{181} Y_{13} - \frac{193}{195} \cos \left(\frac{547}{120} X_{13} + \frac{29}{13} Y_{13} \right) \right) - \frac{93}{167} \cos^4 \left(\frac{1255}{188} X_{13} + \frac{2384}{117} Y_{13} - \frac{94}{143} \cos \left(\frac{326}{107} X_{13} + \frac{849}{145} Y_{13} \right) \right) + \frac{25}{164} \cos^6 \left(\frac{2161}{77} X_{13} + \frac{203}{16} Y_{13} \right)$$

$$X_{14} = \left(X + \frac{68}{161}\right), \quad Y_{14} = \left(Y - \frac{37}{192}\right)$$

$$\tau_{14}(X, Y) = -\frac{50}{179} \cos^2 \left(\frac{514}{31} X_{14} + \frac{915}{193} Y_{14} \right) - \frac{26}{47} \cos^4 \left(\frac{326}{55} X_{14} + \frac{3868}{197} Y_{14} + \frac{139}{141} \cos \left(\frac{45}{67} X_{14} + \frac{117}{28} Y_{14} \right) \right) - \frac{3}{62} \cos^6 \left(\frac{3175}{119} X_{14} + \frac{2325}{187} Y_{14} - \frac{108}{197} \cos \left(\frac{650}{127} X_{14} + \frac{23}{16} Y_{14} \right) \right)$$

$$X_{15} = \left(X + \frac{23}{49}\right), \quad Y_{15} = \left(Y + \frac{5}{129}\right)$$

$$\tau_{15}(X, Y) = \frac{24}{139} \cos^2 \left(\frac{59}{4} X_{15} + \frac{203}{33} Y_{15} - \frac{140}{197} \cos \left(\frac{59}{19} X_{15} + \frac{308}{145} Y_{15} \right) \right) - \frac{15}{197} \cos^4 \left(\frac{1267}{198} X_{15} + \frac{3035}{159} Y_{15} + \frac{19}{66} \cos \left(\frac{120}{127} X_{15} + \frac{723}{148} Y_{15} \right) \right) + \frac{2}{147} \cos^8 \left(\frac{1615}{143} X_{15} + \frac{2589}{71} Y_{15} \right)$$

Master Composition (Recursive Layer Synthesis)

$$H_v^{(0)}(X, Y) = C_0(X, Y, v)$$

$$H_v^1(X, Y) = (1 - W_1(X, Y)) \cdot H_v^0(X, Y) + W_1(X, Y) \cdot C_1(X, Y, v)$$

$$H_v^2(X, Y) = (1 - W_2(X, Y)) \cdot H_v^1(X, Y) + W_2(X, Y) \cdot C_2(X, Y, v)$$

$$H_v^3(X, Y) = (1 - W_3(X, Y)) \cdot H_v^2(X, Y) + W_3(X, Y) \cdot C_3(X, Y, v)$$

$$H_v^4(X, Y) = (1 - W_4(X, Y)) \cdot H_v^3(X, Y) + W_4(X, Y) \cdot C_4(X, Y, v)$$

$$H_v^5(X, Y) = (1 - W_5(X, Y)) \cdot H_v^4(X, Y) + W_5(X, Y) \cdot C_5(X, Y, v)$$

$$H_v^6(X, Y) = (1 - W_6(X, Y)) \cdot H_v^5(X, Y) + W_6(X, Y) \cdot C_6(X, Y, v)$$

$$H_v^7(X, Y) = (1 - W_7(X, Y)) \cdot H_v^6(X, Y) + W_7(X, Y) \cdot C_7(X, Y, v)$$

$$H_v^8(X, Y) = (1 - W_8(X, Y)) \cdot H_v^7(X, Y) + W_8(X, Y) \cdot C_8(X, Y, v)$$

$$H_v^9(X, Y) = (1 - W_9(X, Y)) \cdot H_v^8(X, Y) + W_9(X, Y) \cdot C_9(X, Y, v)$$

$$H_v^{10}(X, Y) = (1 - W_{10}(X, Y)) \cdot H_v^9(X, Y) + W_{10}(X, Y) \cdot C_{10}(X, Y, v)$$

$$H_v^{11}(X, Y) = (1 - W_{11}(X, Y)) \cdot H_v^{10}(X, Y) + W_{11}(X, Y) \cdot C_{11}(X, Y, v)$$

$$H_v^{12}(X, Y) = (1 - W_{12}(X, Y)) \cdot H_v^{11}(X, Y) + W_{12}(X, Y) \cdot C_{12}(X, Y, v)$$

$$H_v^{13}(X, Y) = (1 - W_{13}(X, Y)) \cdot H_v^{12}(X, Y) + W_{13}(X, Y) \cdot C_{13}(X, Y, v)$$

$$H_v^{14}(X, Y) = (1 - W_{14}(X, Y)) \cdot H_v^{13}(X, Y) + W_{14}(X, Y) \cdot C_{14}(X, Y, v)$$

$$H_v^{15}(X, Y) = (1 - W_{15}(X, Y)) \cdot H_v^{14}(X, Y) + W_{15}(X, Y) \cdot C_{15}(X, Y, v)$$

$$H_v(X, Y) = H_v^{15}(X, Y)$$

Final RGB Channel Evaluation

$$\text{Color}(x, y) = (F(H_0(X, Y)), F(H_1(X, Y)), F(H_2(X, Y)))$$

where $F(x) = \left[255 e^{-e^{-1000x}} |x| e^{-e^{1000(x-1)}} \right]$ performs exact differentiable color clamping to $[0, 255]$.